

Pen Test Pocket

Final Report

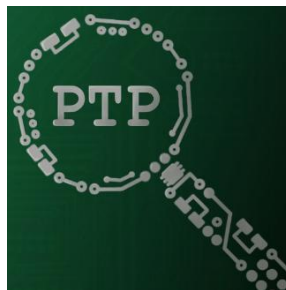
Dayton Dekam - CrowdStrike



Mentor:

Ananth Jillepalli, Scholarly Assistant Professor

PTP Adelson



Solo Project

Zachary Adelson

CptS 432 Cybersecurity Capstone Project

Fall 2025

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I. Introduction

I.1. Project Introduction

As a student and upcoming professional in Cybersecurity, it is essential to always and consistently be upskilling, by learning new tools and skills. One of the biggest hurdles I have found when it comes to learning and using tools is remembering all the possible input flags for command line-based interaction. The **PenTestPocket (PTP)** solves the struggles that are associated with learning tools, and then not only simplifies the process but improves on the experience with the addition of new features.

To accomplish these goals, I created GUI based toolkit rather than the started CLI based ones, which allows users to fully interact with and use common pen testing tools such as Nmap, NetCat, etc. While there are many tools available as web applications that allow for tools similar to **PTP**, such as [nmap.online](https://nmap.org) or [RapidTables Conversion tools](#), that can each individually fulfill the goal of the **PTP** in making these tools and these tasks easy to understand and view without requiring working with a CLI. **PTP** congregates all these tools into a single tool, so no jumping around between websites and tools, everything you will need will always be right in front of you.

Some of the additional features that **PTP** has to offer are logging of all actions and test results for later review, pinning of tools to the homepage of the application for ease of access and use, and a built-in written guide for all the tools in question. These features will enable larger scale tests to be automatically documented, and all that would be required of the end user is reformatting and copying from the user log files which are encrypted within the file but will be able to viewed in clear text within **PTP** so only authorized users can access the log files for confidentiality purposes. We will describe how **PTP** maintains the CIA triad later in this report.

1. My primary client is Dayton Dekam from CrowdStrike, where he agreed to represent as the professional sponsor of the project, but this project is directly affiliated with CrowdStrike. My mentor and professor Ananth Jillepalli serves as the main point of communication for the project. The final project will highlight the skills and capabilities offered by **PTP**

2. Dayton Dekam, CrowdStrike, daytondekam@crowdstrike.com

3. According to [CrowdStrike's official website](#) "CrowdStrike is a global cybersecurity leader with an advanced cloud-native platform for **protecting endpoints, cloud workloads, identities and data**". Dayton has worked with CrowdStrike for over 3 years and currently as a Cyber Analyst 2

I.2. Project Developer Bio

Zachary Adelson is a Senior at Washington State University and will graduate with his Bachelor of Science degree in Cybersecurity, a minor in mathematics, and the CySER CAE-CO Fundamentals Certificate in May of 2026. Zachary has always had an interest in technology and aims to make technology the core focus of his professional career. Prior to PTP, Zachary has participated in multiple CTF and Hackathon competitions, has held two internships at two very distinct companies where he worked as an IT technician and a Technology Risk Assurance auditor for each respective company. As this is a solo project, under the guidance of Ananth Jillepalli and the sponsor backing of Dayton Dekam, Zachary is responsible for all aspects of this project, from GUI design, creation, and code development.

II. Project Requirements Specification

From your revised “Project Requirements Specification” document include the following:

II.1. Project Stakeholders

The stakeholders are CrowdStrike, my professor, and myself, but are not restricted to them.

Potential stakeholders can include any cybersecurity professional. As this tool is geared towards enabling an easy and more efficient testing workflow, **PTP** is the perfect project for anyone looking to break into cybersecurity and streamline their process flow.

All stakeholders involved would benefit from having a clear and concise way of performing penetration tests and that all user logging can be used for future personal documentation, which is a key part of any sort of reporting.

II.2. Use Cases

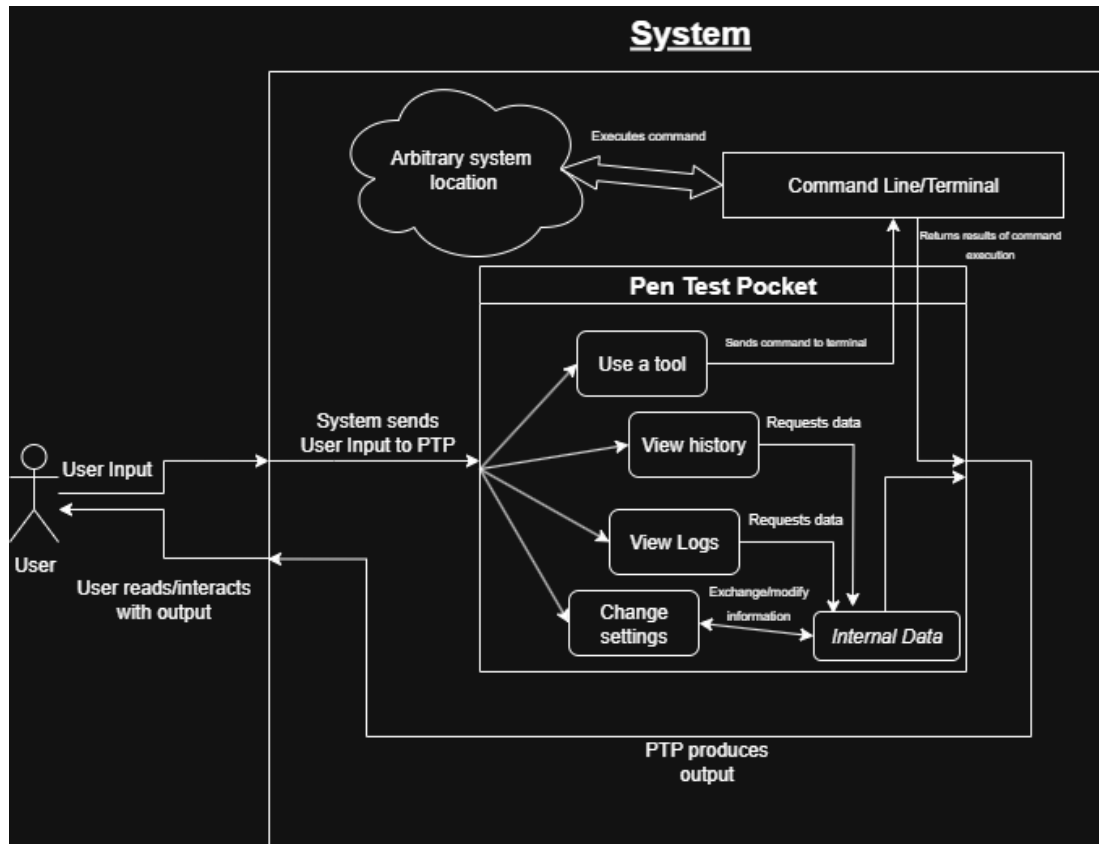


Figure 1: PTP UML Diagram

User Input

<i>Pre-condition</i>	- PTP is open and running
<i>Post-condition</i>	- User input processed - PTP can now act on that input - Events logged
<i>Intended Path</i>	1. Open PTP 2. Select any user interface 3. Interact with said interface to do an action
<i>Alternative Path</i>	
<i>Related Requirements</i>	- Use a tool - View history - View Logs - Change settings

Use a tool

<i>Pre-condition</i>	- User has defined what tool is being called - User has defined all necessary parameters and flag(s)
<i>Post-condition</i>	- Results of tool use received and returned - Tool results logged
<i>Intended Path</i>	1. Selection of tool 2. Passed in parameters and flag(s)
<i>Alternative Path</i>	I. Integrated terminal II. Use tool as if it were CLI based
<i>Related Requirements</i>	- Systems terminal - Systems related resources

View History

<i>Pre-condition</i>	- A tool(s) has been executed
<i>Post-condition</i>	- Specific tool history information displayed
<i>Intended Path</i>	1. Selection of tool within history box
<i>Alternative Path</i>	
<i>Related Requirements</i>	- Internal PTP data

View Logs

<i>Pre-condition</i>	
<i>Post-condition</i>	- All logs are displayed to the user
<i>Intended Path</i>	1. Log button selected
<i>Alternative Path</i>	
<i>Related Requirements</i>	- Internal PTP data

Change settings

<i>Pre-condition</i>	- Settings configurations exist
<i>Post-condition</i>	- Any changes are applied if save is selected
<i>Intended Path</i>	1. Go to settings button
<i>Alternative Path</i>	
<i>Related Requirements</i>	- Internal PTP data

II.3. Functional Requirements

1. Graphical User Interface

<i>Description</i>	The PTP application is designed around having a Graphical User Interface (GUI) for the users to interact with. To be able to achieve this goal I found a python3 based GUI stack, of which I am using PyQt5
<i>Dependencies</i>	Python3, PyQt5, related python3 libraries, all integrated tools (Nmap, hashing tools, etc.)
<i>Categorization</i>	Code ; as defined on GitHub roadmap

2. Tool Integration

<i>Description</i>	As the pen testing tools are used within the application itself and not in external location such as CLI, it is crucial to the success of the application that the tools be seamlessly incorporated into the PTP application. Tools should be designed to be implemented in a way that allows for ease of modification, removal, or additions
<i>Dependencies</i>	Nmap, netcat, smbclient, snmpwalk, onesixtone, gobuster, curl, whatweb, Metasploit, searchsploit, hashcat, john the ripper, encoding/decoding (binary, hexadecimal, decimal, base64, ascii, utf), SHA-1, SHA-256, MD5, CRC-32, RIPEMD-160, B-crypt
<i>Categorization</i>	Code ; as defined on GitHub roadmap

3. Terminal Integration

<i>Description</i>	PTP intends to have an integrated terminal so that any terminal-based commands can be run within PTP without the need of opening an independent terminal to run one or two simple commands. The integrated terminal will communicate with the systems terminal to get real and accurate execute results
<i>Dependencies</i>	BASH, PowerShell
<i>Categorization</i>	Code ; as defined on GitHub roadmap

II.4. Non-Functional Requirements

Succinct GUI:

- The GUI should be simple to understand. There should be little to no learning curve required for utilizing PTP, as that would detract from the overall goal of PTP, shifting learning the tools to learning how to use the tool to simplify the tools.

Ease of Operation:

- Accessing and using any part of PTP should be able to be achieved without much effort. All tool operations should be streamlined such that you'd only need to read the text presented about the tool in order to figure out how to use it.

All Encompassing Needs Met:

- PTP will have every aspect of execution of workflow integrated within the tool itself, including but not limited to; Tool help, tool execution, secure note taking, and secure log files.

Maintaining CIA:

- PTP will maintain all three sectors of CIA in the following ways
 - *Confidentiality* – Using encryption techniques to prevent our user log or note data isn't stored and plain text and cannot be exfiltrated easily by potential threat actors.
 - *Integrity* – By making use of hashing, we can ensure that data within PTP isn't be changed without intention or permission to do so.
 - *Availability* – PTP will ensure that the application is always available when needed and all elements are accessible, and by maintain our confidentiality and integrity we can be sure that our program will always be accessible.

III. Software Design - From Solution Approach

III.1. Architecture Design

PTP was designed with three aspects in mind, security, simplicity, and expandability

III.1.1. Overview

PTP has security as its forefront as the app is designed to test security, it will make sure its threat resilient with the CIA scheme. It is simple in its end user design such that users will be able to use the tool without any additional hassles other than reading the text on the screen in front of them. Lastly, PTP is built to be easily able to add additional tools and alter the code for user cases

III.1.2. Subsystem Decomposition

I broke PTP into a few separate parts; Main page, tool page, settings tab, change log, help window, terminal screen, and user log.

- Main Page
The main page includes all the pinned tools, connections to all other tabs, and history of tool executions
- Tool Page
The tool page will pop up whenever a tool is selected, and all its associated attributes will be available for customization based on user needs
- Settings Tab
The settings tab deals with everything regarding customization of the application including theme, sound settings, terminal customization
- Change Log
The change log will keep track of and display all changes and additions made to PTP by me throughout the course of its development
- Help Window
The help window will hold all information regarding use of PTP including navigation of the GUI, individual tool specifications, and how the app works
- Terminal Screen
The terminal screen is an integrated section of PTP that acts like a real terminal but in the background only acts as a middleman between the user and the system terminal
- User Log
The user log is where all user events are tracked and stored for later viewing, these can range anywhere from user launching the application, to exactly what tool was run and when

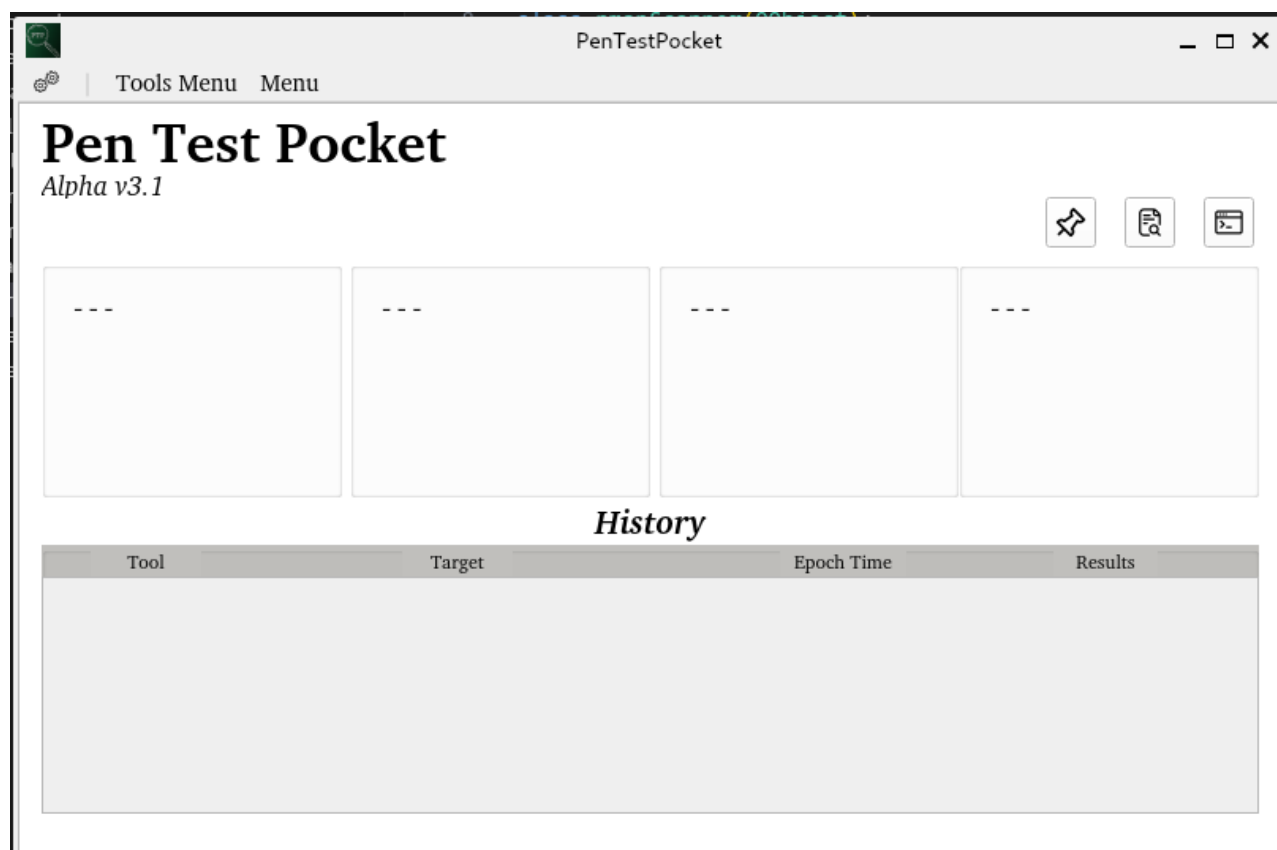
III.2. Data design

Within PTP I use lists, dictionaries, and tuples to store all temporary and persistent information, as well as txt and JSON files for all additional storage that needs to persist between sessions.

III.3. User Interface Design

The GUI has multiple sections as mentioned above: Main page, tool page, settings tab, change log, help window, terminal screen, and user log. All tools are currently in progress, but each will be documented at its current stage below:

Main Page



Tool Page

PenTestPocket

Nmap

Run Nmap

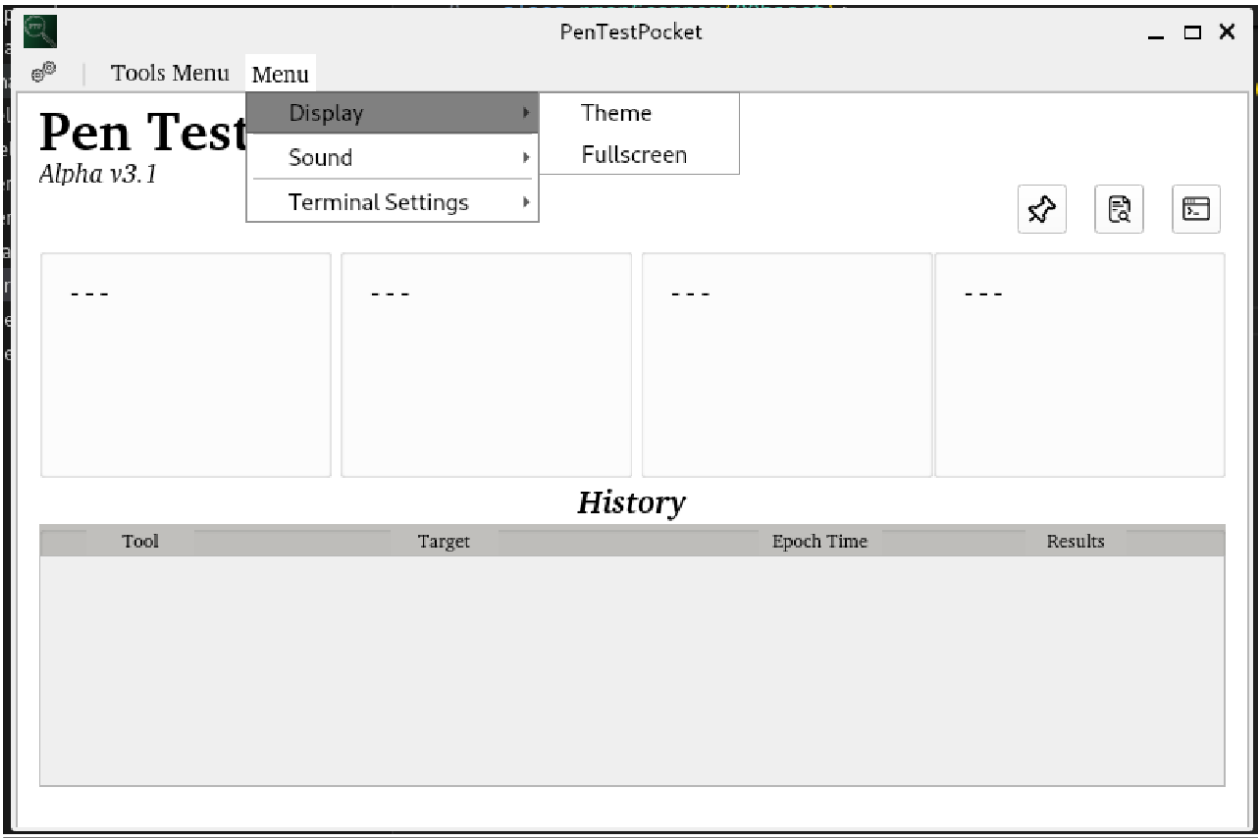
Enter Target...?

☐ Exclude Target

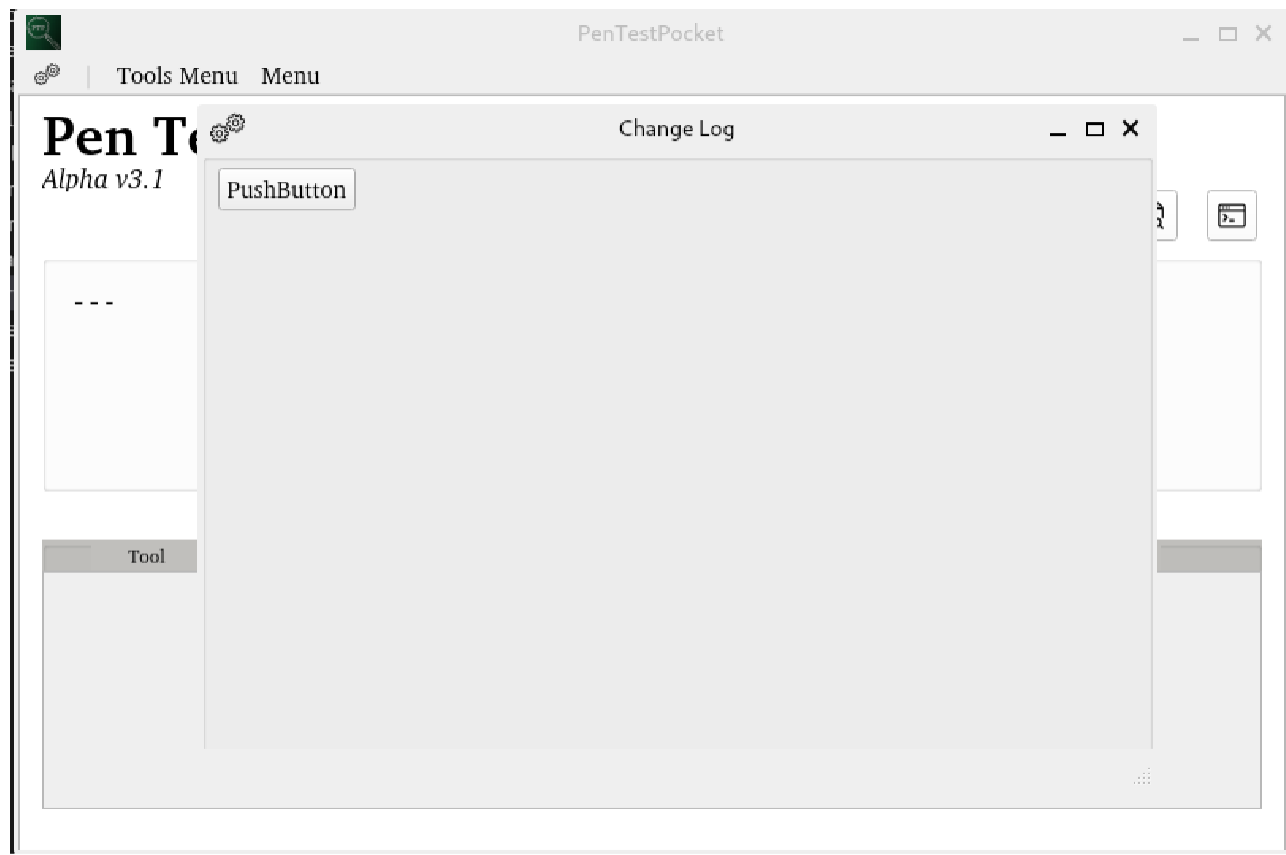
No Specific Scan Type

No Specific OS Scan

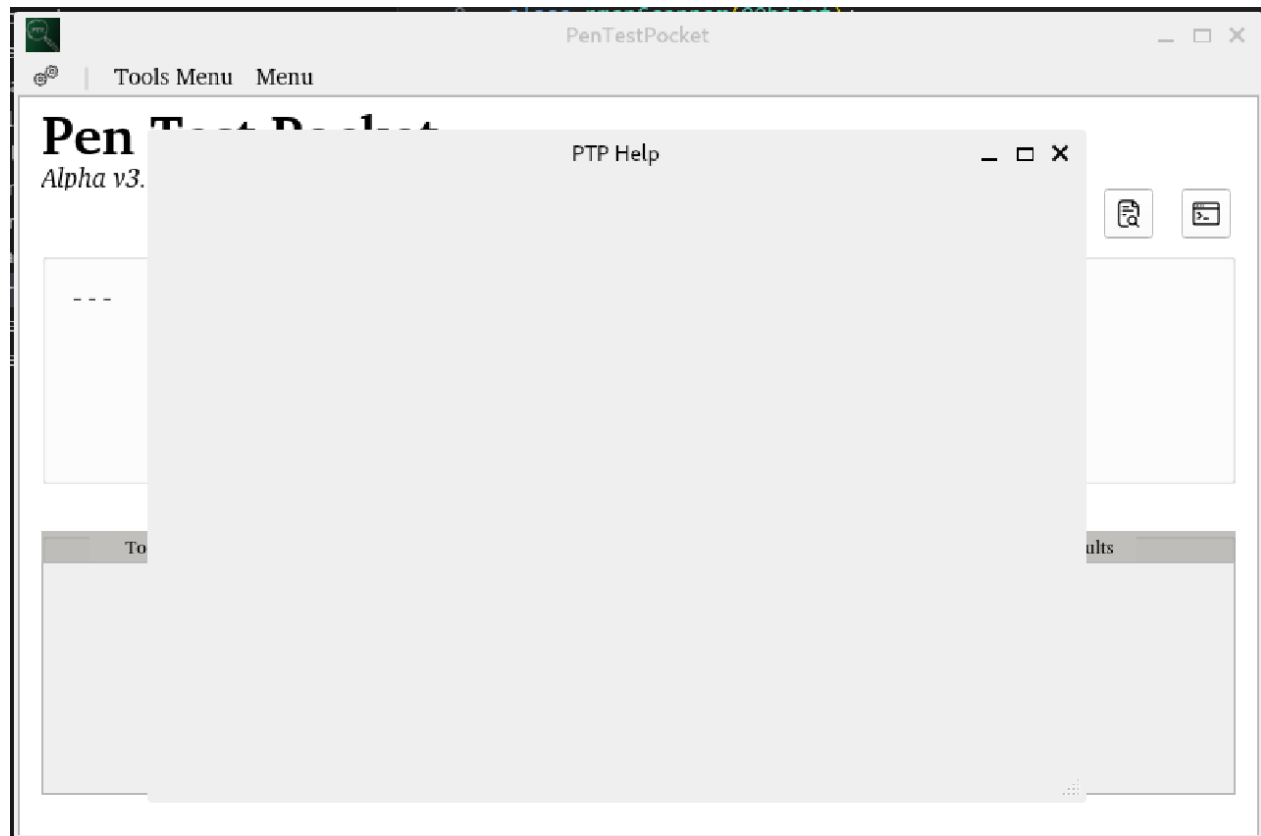
Settings Tab



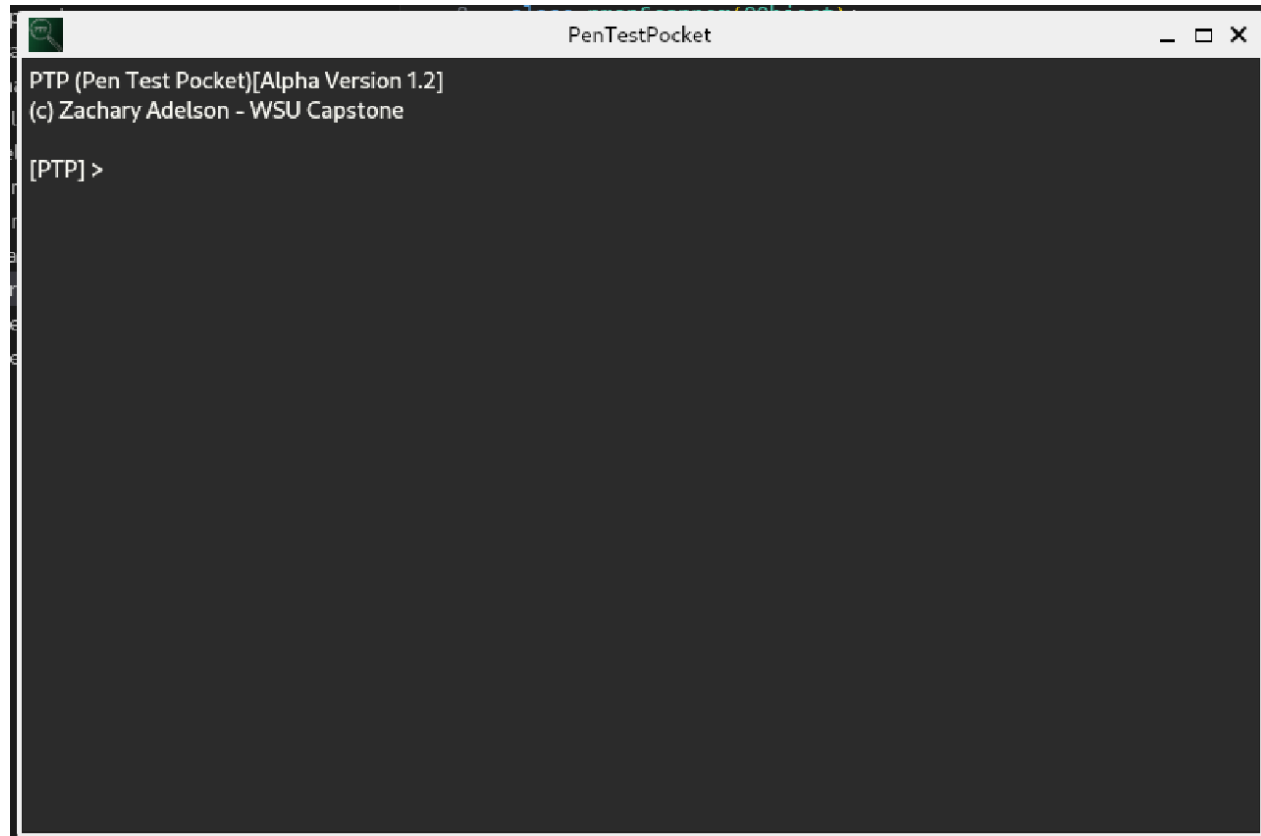
Change Log



Help Window



Terminal Screen



User Log



III.4. Constraints

This project has faced numerous constraints in terms of its development, particularly balancing classes and personal work. However, the project itself faced constraints; The path that I had to go down for developing using PyQt5. I faced restrictions with ensuring that the tool was focused on ease of use over power and capability. Ensuring that everything remains simple to setup, use, and operate.

IV. Test Case Specifications and Results

IV.1. Testing Inputs

Test 1 – Nmap comparisons

- a. Test Input – *nmap 127.0.0.1*
- b. Test Input – *nmap 8.8.8.8*
- c. Test Input – *nmap -exclude 8.8.8.8*

Test 2 – Ncat comparisons [Not Completed Enough For Testing]

- a. Test Input –
- b. Test Input –
- c. Test Input –

Test 3 – Hashcat comparisons [Not Completed Enough For Testing]

- a. Test Input –
- b. Test Input –
- c. Test Input –

Test 4 – curl comparisons

- a. Test Input – *curl https://google.com/*
- b. Test Input – *curl https://bing.com/*
- c. Test Input – *curl https://yahoo.com/*

Test 5 – Data conversion comparisons

- a. Test Input – *Decimal -> Hexadecimal 321*
- b. Test Input – *Binary -> Text 01001000 01100101 01101100 01101100
01101111 00100000 01000011 01100001 01110000 01110011 01110100
01101111 01101110 01100101*
- c. Test Input – *Text -> Base64 Goodnight Capstone*

Test 6 – Hashing comparisons

- a. Test Input – *SHA-1 Hello Capstone*
- b. Test Input – *MD5 Hello Capstone*
- c. Test Input – *Shake256 Hello Capstone*

IV.2. Environment Requirements

To support testing, PyQt5 alongside Python3 must be installed, and as the project progresses all additional installed tools will also need to be present, all requirement installation commands are listed on GitHub readme page.

Additional tools

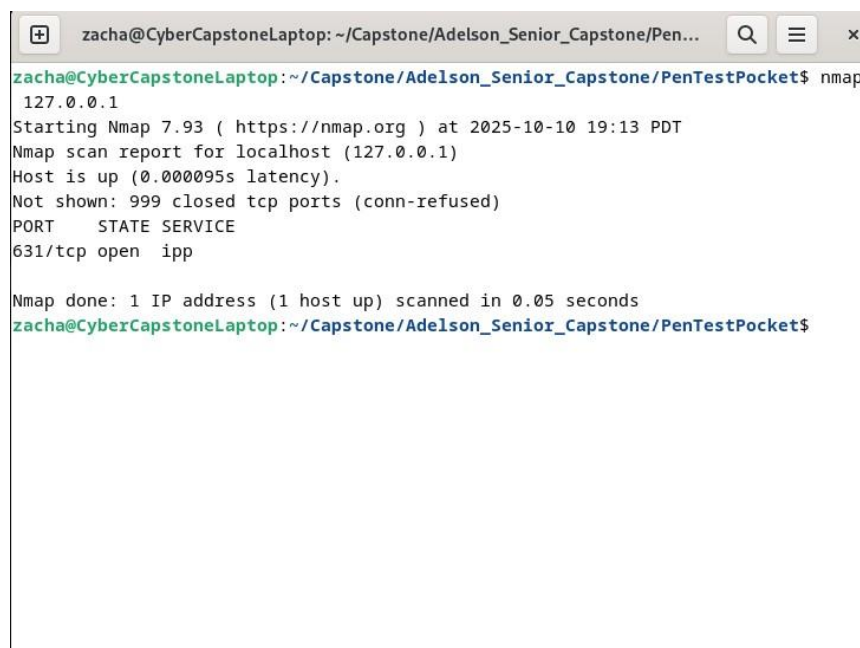
- Nmap
- Ncat
- Hashcat
- Curl

IV.3. Test Results

Test 1 Results – Nmap Comparisons

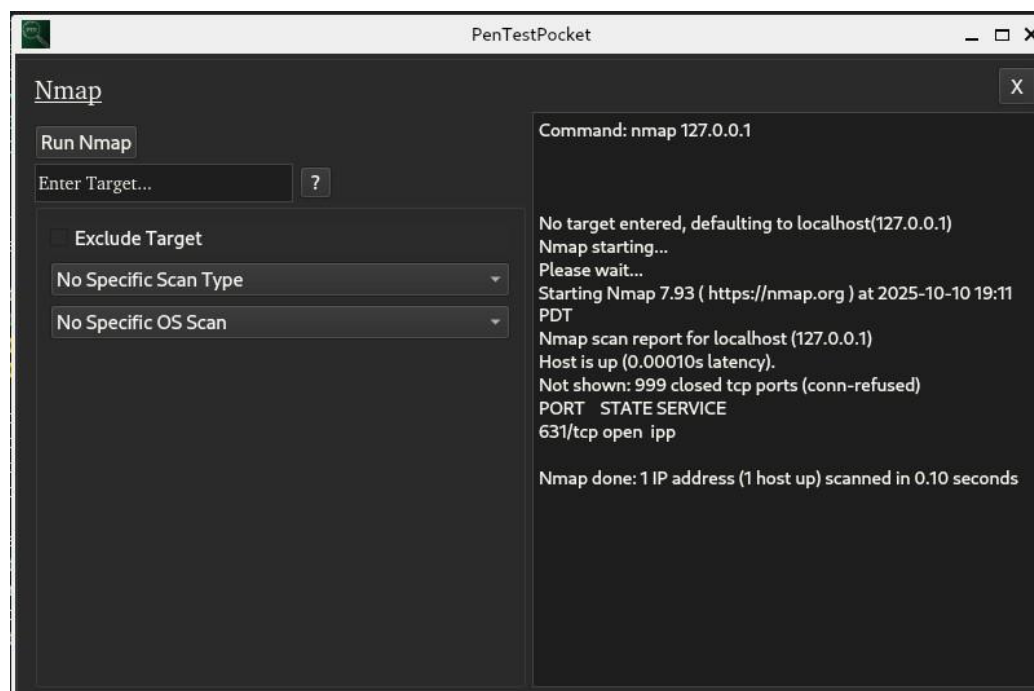
1a.

Base Case



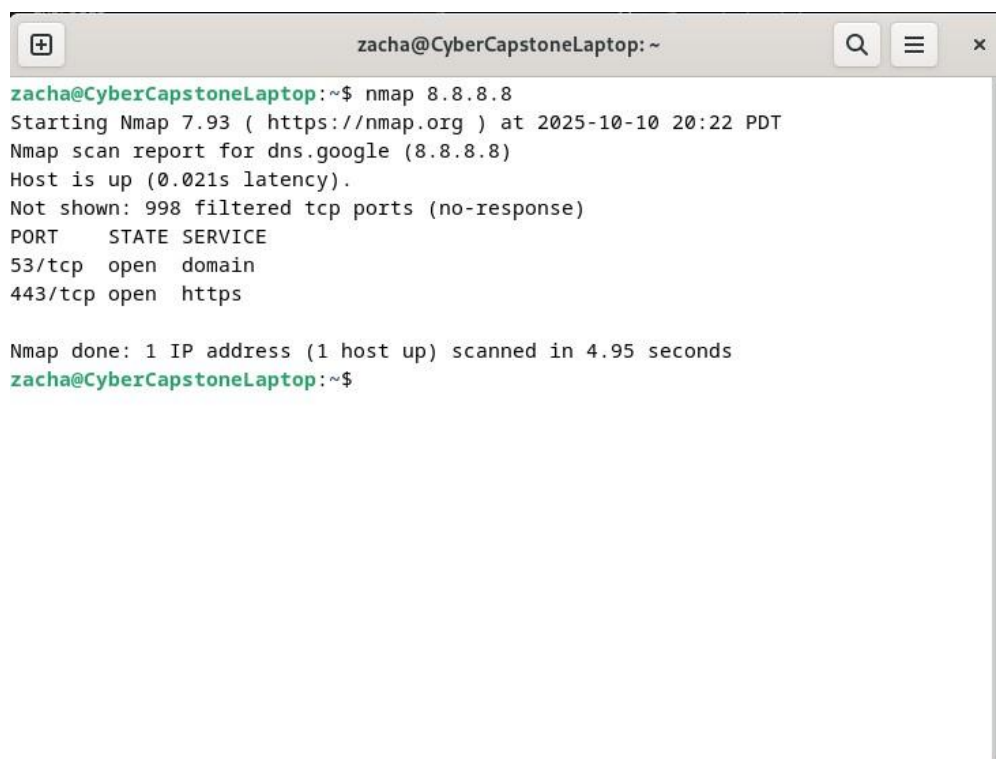
```
zacha@CyberCapstoneLaptop: ~/Capstone/Adelson_Senior_Capstone/PenTestPocket$ nmap 127.0.0.1
Starting Nmap 7.93 ( https://nmap.org ) at 2025-10-10 19:13 PDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000095s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT      STATE SERVICE
631/tcp   open ipp
Nmap done: 1 IP address (1 host up) scanned in 0.05 seconds
zacha@CyberCapstoneLaptop: ~/Capstone/Adelson_Senior_Capstone/PenTestPocket$
```

Test Case



1b.

Base Case

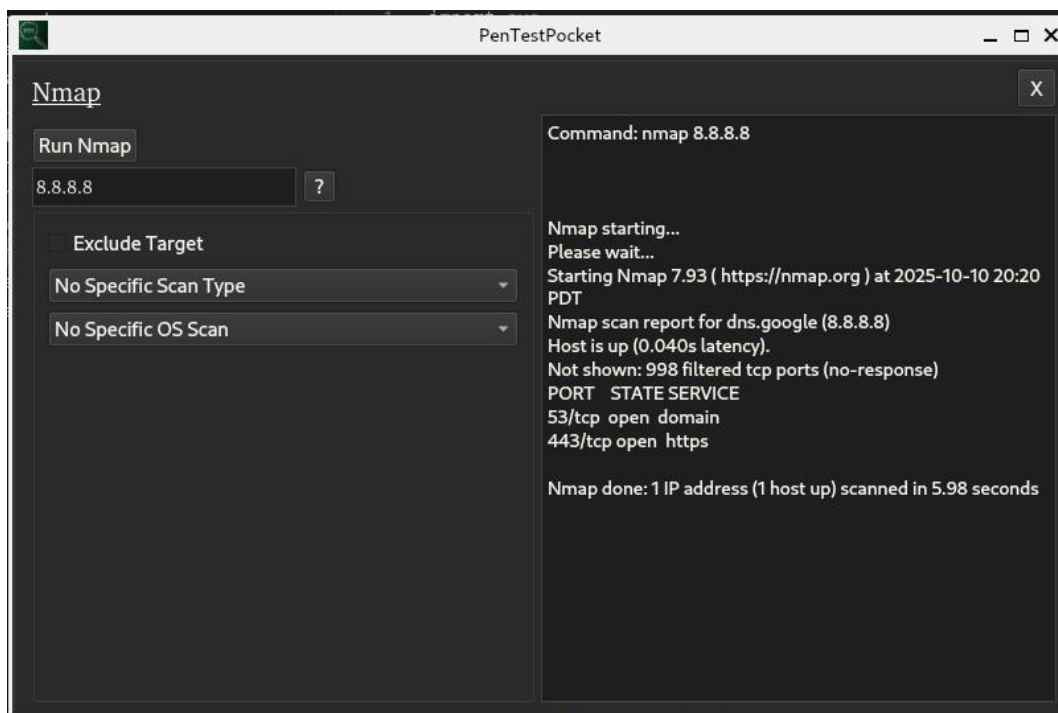


A terminal window titled 'zacha@CyberCapstoneLaptop: ~' showing the output of an nmap scan. The command 'nmap 8.8.8.8' is entered. The output includes the nmap version (7.93), the target (8.8.8.8), the scan time (2025-10-10 20:22 PDT), and the results of the scan. The host is up with a latency of 0.021s. The scan shows two open ports: 53/tcp (domain) and 443/tcp (https). The scan is completed in 4.95 seconds.

```
zacha@CyberCapstoneLaptop:~$ nmap 8.8.8.8
Starting Nmap 7.93 ( https://nmap.org ) at 2025-10-10 20:22 PDT
Nmap scan report for dns.google (8.8.8.8)
Host is up (0.021s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 4.95 seconds
zacha@CyberCapstoneLaptop:~$
```

Test Case



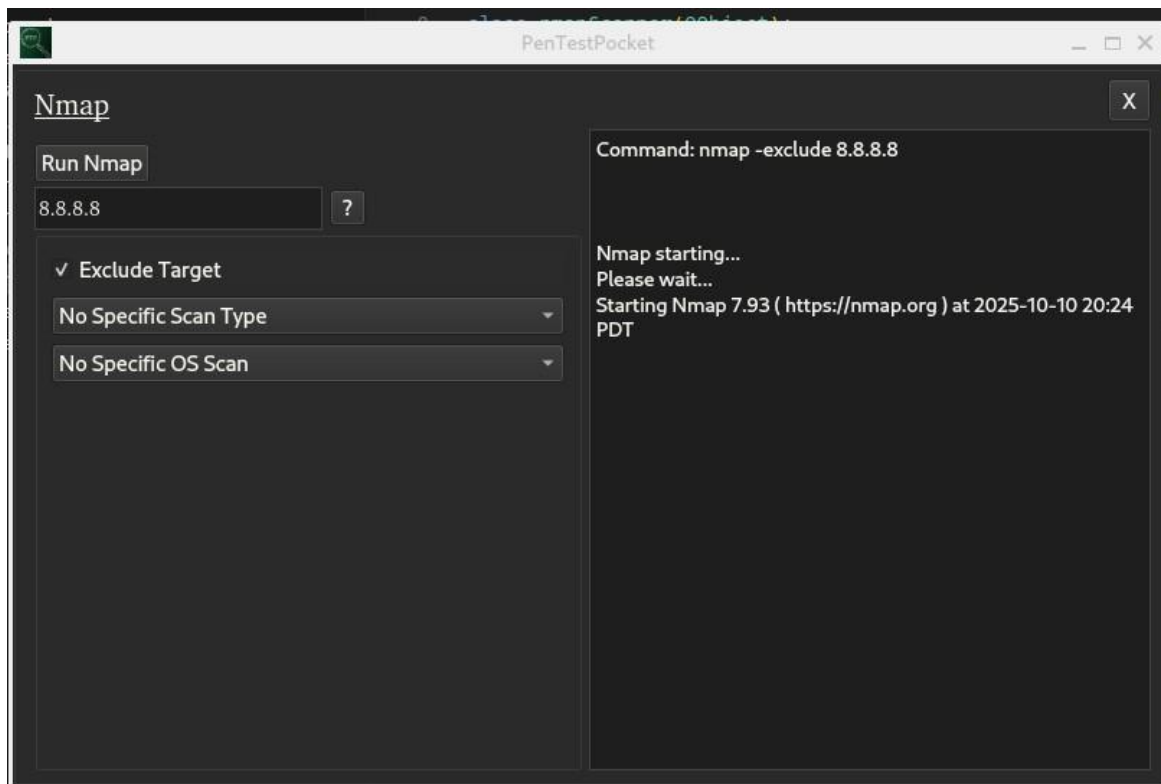
1c.

Base Case

A terminal window titled 'zacha@CyberCapstoneLaptop: ~' with search, menu, and close icons. It shows the execution of the command 'nmap -exclude 8.8.8.8'. The output indicates that Nmap 7.93 started at 2025-10-10 20:24 PDT, issued a warning about no targets specified, and completed scanning 0 IP addresses in 0.03 seconds.

```
zacha@CyberCapstoneLaptop:~$ nmap -exclude 8.8.8.8
Starting Nmap 7.93 ( https://nmap.org ) at 2025-10-10 20:24 PDT
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.03 seconds
zacha@CyberCapstoneLaptop:~$
```

Test Case



Remaining tests linked below:

https://github.com/zachA214/Adelson_Senior_Capstone/blob/main/Documentation/TestingResultsProjectReport/PTP_TestingResults.pdf

IV.4. Projects and Tools used

<u>Tool/library/framework</u>	<u>Designed Usage(s)</u>
PyQt5	Designing GUI and allowing for simple, cross platform application creation and development
	Intercommunication between our modules

<u>Languages Used in Project</u>			
Python3	BASH shell	PowerShell	LaTeX
JSON			

V. Description of Final Prototype

PenTestPocket, PTP, is a penetration testing toolkit that takes all the commonly seen testing tools and bundles them into one single product and adding additional benefits to give reason to swap from the traditional way of testing. The toolkit is built using the PyQt5 framework to give a GUI so that beginners to the scene are not required to memorize every command that is used but professionals also are not slowed down by the swap. The toolkit features advanced user logging alongside tool pinning for ease of use. Multiple tools have been integrated and are shown in sprint testing videos. Installation is designed to be streamlined and easy to do, only requiring downloading of the zip file and one terminal download command that is listed in the installation requirements on GitHub.

VI. Social Responsibility and Broader Impacts

Social Responsibility: When developing this project, the judgements I made were in line with an ethical principle such that every action used by PTP would be in accordance with the owners of the systems being tested on, in a professional or academic situation with no malicious intent. These judgements are entirely in line with my professional responsibilities as I am working to ensure that all penetration testing cases are not without consent and understanding of the scope.

Broader Impacts: My work with this capstone project aligns with WSU EECS's goals by presenting a platform to allow all who are interested in penetration testing and cybersecurity to have a simple to understand gateway into the field while still making the penetration testing experience more streamlined for professionals. My project has a thorough documentation from initial setup to code deep dives alongside of having the GPL-3.0 license to support open source missions and bring this learning to everyone.

VII. Product Delivery Status

The project has been presented to Ananth Jillepalli on 12/5/2025 over zoom, in lieu of Dayton who was unavailable. It was demonstrated and recorded.

The repository is linked [here](#); All requirements to install the project are linked within the GitHub only Debian 12 is tested and working, however.

VIII. Conclusions and Future Work

VIII.1. Limitations and Recommendations

The limitations of PTP are that it is currently restricted exclusively to Debian based Linux distributions, it is entirely reliant on having the tools (such as nmap, ncat, etc.) installed to function properly, and any bugs that occur do not fail safely causing the program to crash at any stage. Moving forward with this project my recommendations would be to enforce the tools to be downloaded to start the program, develop to work on multiple distributions and operating systems, and include more try catch statements so that if an error occurs it halts the process where the error occurred but doesn't crash PTP.

VIII.2. Future Work

Future Work consists of finishing all the remaining tools that were not implemented along with increasing the breadth of all tools such that you are not missing any capabilities when moving over to PTP. Tool outputs need to be stored so that they can be viewed or managed at any later time, then hashing and encryption needs to be implemented for all tool outputs.

IX. Acknowledgements

I would like to acknowledge and thank my professor and mentor Ananth Jillepalli for supporting my project and collaborating with me to get this project off the ground running. PTP was independently designed and thought up by me, and Ananth Jillepalli worked tirelessly to ensure I had approval from the department and that my project would make sense within the scope of a capstone project. I would also like to thank my professional sponsor Dayton Dekam from CrowdStrike for believing in my idea enough to function as the sponsor when the university was unable to gather sponsors for the senior capstone project course. Lastly, I would like to thank the departments of EECS (Electrical Engineering and Computer Science) and VCEA (Voiland College of Engineering and Architecture) for approving my project and allowing me to bring my vision into reality with this semester long project.

X. Glossary

PTP - PenTestPocket

CLI – Command Line Interface

PyQT/Qt – Qt framework for GUI

EECS – Electrical Engineering and Computer Science

VCEA – Voiland College of Engineering and Architecture

XI. References

<https://www.reddit.com/r/learnpython/comments/94wszv/comment/e3qt1ax/>

<https://youtube.com/watch?v=pAifkTKvC84>

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<https://www.w3schools.com/python/python-json.asp>

XII. Appendix A - Example Testing Strategy Reporting

When testing the goal is to ensure that buttons and code are executing how we expect, and that tools are returning information as the original tool does.

When testing tools, we will run the command in the terminal, screenshot the results, and run the command in PTP screenshot those results, and compare, and if they are identical, we know our test is successful

When testing code, we describe exactly what the code is intended to do, and then run those steps, executing the code, and document the output with screenshots

XIII. Appendix B - Project Management

Every week since this is a solo project is just independent work on the Capstone assignment. On Thursdays before the weekly mentor meeting, I will add new issues for the coming week and upload my time tracking excel sheet, which is used to tabulate all the time I have spent on the Capstone project each day, by categorization.

For project planning, I use GitHub issues and the GitHub roadmap grouped into 4 distinct groups; Logistics, Code, Meetings, and Reporting.

Logistics is for everything project management wise such as blocking time, planning next steps, etc.

Code is for the time spent working on the actual application side, including all time spent developing and testing PTP.

Meetings are reserved for the time spent during mentor meetings, which on a standard week is just 30 minutes on Thursdays.

Reporting is for any write-ups and documentation, including sprint and project reports; This is also meant for any instructions or documentation needed to view and use PTP.